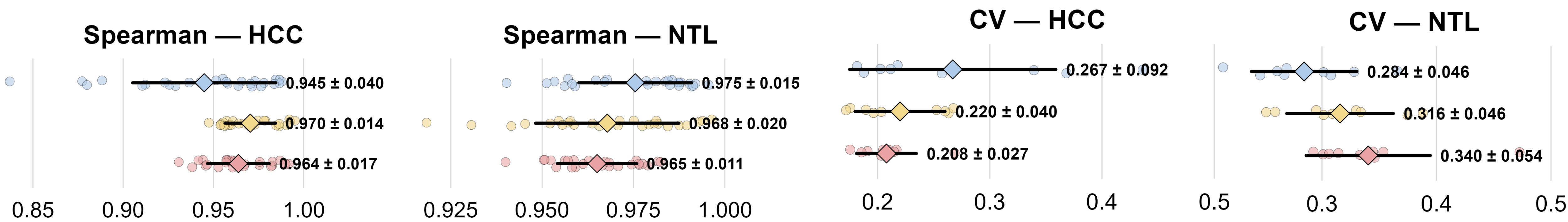
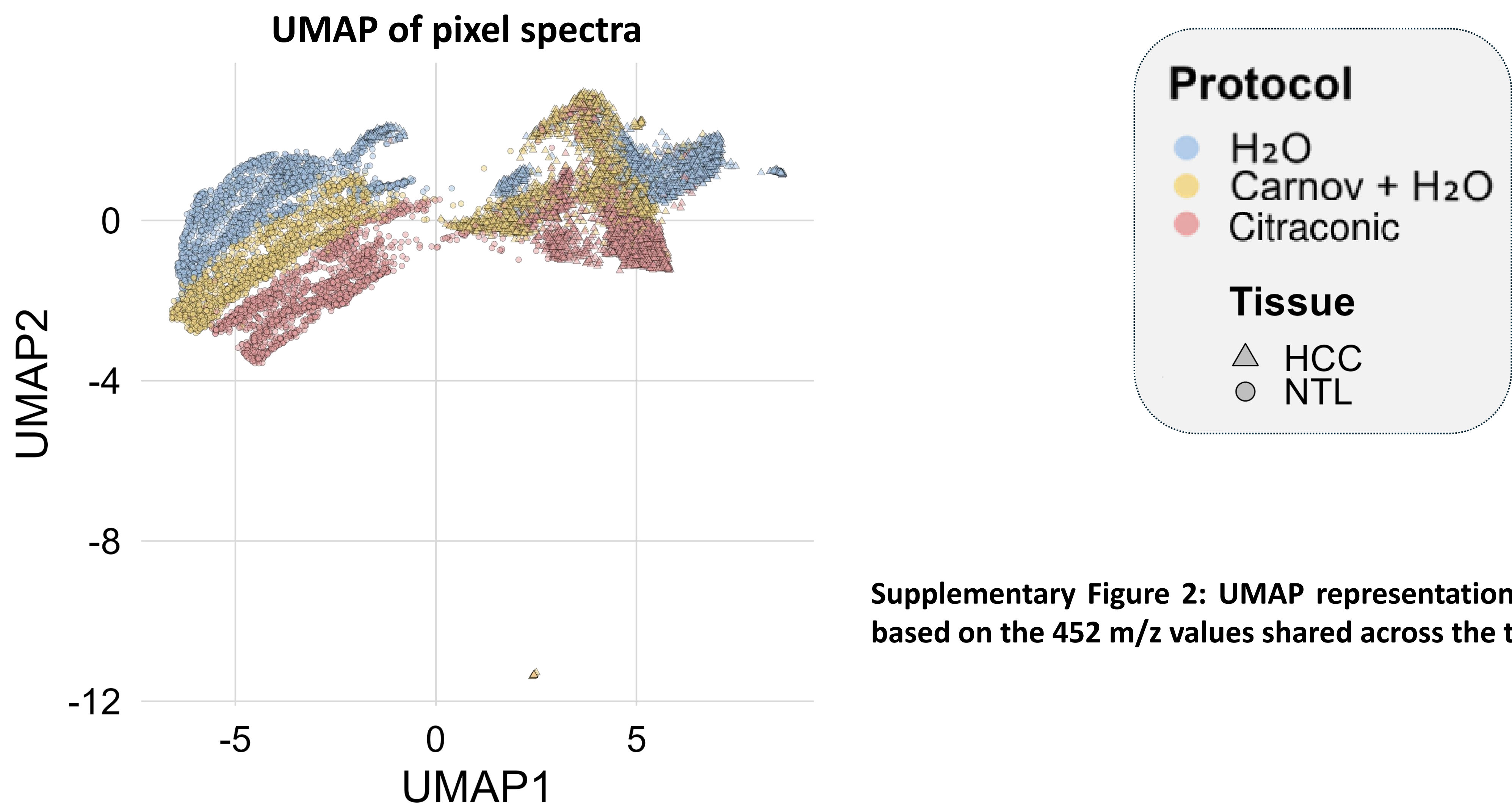


Supplementary figures



Supplementary Figure 1: Spectral variability and reproducibility across selected protocols. Intra-ROI CV was calculated from the 300 most intense m/z features per ROI, whereas Spearman correlations were computed between full harmonized mean ROI spectra. Lower CV and higher correlations indicate better analytical performance.



Supplementary Figure 2: UMAP representation of individual pixel spectra based on the 452 m/z values shared across the three protocols.